Sample exam questions

Foundations of Operations Research

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Problem 1

A company production plant consists of 6 buildings that are connected via 12 walking roads. The distances among the 6 buildings are reported in the following table:

	1	2	3	4	5	6
1	-	3	2	4	4	5
2	3	-	-	-	2	7
3	2	-	-	1	-	2
4	4	-	1	-	10	10
5	4	2	-	10	-	12
6	5	7	2	10	12	-

where "-" indicates the absence of a direct rooad. The only exit is located in building number 6: in case of emergency all employees have to evacuate the area by reaching building number 6.

The security manager of the company must determine the path to be followed by the employees so that the area is evacuated as fast as possibile. Clearly the time needed for evacuation is proportional to the length of the evacuation path and the road lengths are symmetric. Since the walking roads are sufficiently large with respect to the number of employees, the capacities of the roads can be neglected.

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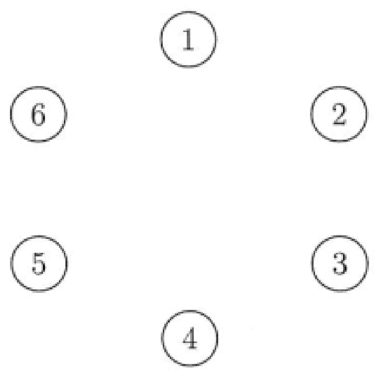
a) **Draw** the **graph** where each node corresponds to a building and each edge corresponds to a walking road (symmetric length).



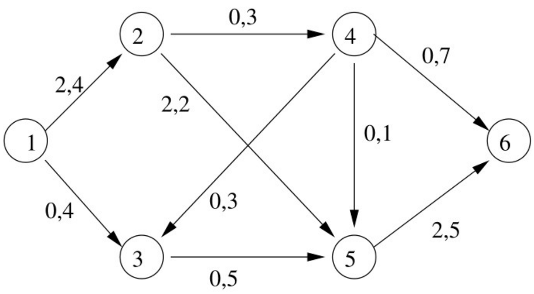
b) The above problem amounts to which classical graph optimization problem? **Precisely describe** the problem, carefully specifying the inputs and the outputs.

c) **Describe** the **most efficient algorithm** studied in the course that can be used to solve the above instance of the problem.

d) **Apply** this algorithm to the instance under consideration, report all the steps and indicate the optimal solution found (the optimal evacuation plan).

Problem 2

Consider the following network where the **first** number associated to each arc (link) corresponds to the **initial** amount of flow (in Gbit/s) crossing it and the **second** number to the arc **capacity** (in Gbit/s).



Determine a feasible flow of maximum value from node 1 to node 6, starting from the given feasible flow of value 2.

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a) Which algorithm can be used? Apply it starting from the given feasible flow of value 2 and reporting all the intermediate steps.

b) Draw the network and indicate the maximum feasible flow you have found, specifying the amount of flow through each arc and the corresponding total value of the maximum feasible flow.

c) Indicate a cut separating the source (node 1) from the sink (node 6) of minimum capacity on the network of point (b).

d) Consider a generic network with n nodes and m arcs. What is the worst-case complexity of the above algorithm when arbitrary augmenting paths are selected at each iteration? Provide a detailed explanation.

Problem 3

Consider the project of assembling an electronic device by appropriately integrating three components. The project is composed of 7 activities with the duration and predecessors reported in the following table:

Activity	Duration (days)	Predecessors
A	3	-
B	5	-
C	6	-
D	4	A
E	3	A, B, D
F	4	B, C, D
G	2	E, F

The activities A, B and C correspond to the detailed testing procedures of, respectively, components C_1, C_2 and C_3. The activity D consists in testing and connecting an output modulator for component C_1, while the activities E and F consist in testing and setting up integrators for the outputs of some of the components. Finally, the activity G corresponds to the test and quality control of the overall electronic device.

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a) Draw the directed graph (with the arcs associated to the activities) corresponding to the above project and indicate the earliest and latest times for the event associated to each node.

b) Indicate the overall minimum completion time for the project.

c) Determine the slackness of each activity.

d) Indentify the critical activities.

e) What is the order of complexity of the algorithm used? Provide a detailed explanation.

Problem 4

Given the following linear programming problem:

$$\begin{aligned} \max z = 2x_1 + x_2 & \\ x_1 + x_2 &\leq 4 \\ 3x_1 + x_2 &\leq 9 \\ x_1, x_2 &\geq 0 \end{aligned}$$

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a) Express the problem in standard form.

b) Perform <u>a single iteration</u> of the Simplex algorithm with <u>Bland's rule</u>. Report all the steps, including the initial tableau and the resulting tableau. Indicate the corresponding basic feasible solution and its objective function value.

c) The basic feasible solution found is optimal? Motivate your answer.

d) Write the dual problem of the original linear program.

e) Solve the dual problem graphically and indicate the optimal solution found and its objective function value.

f) Derive the optimal solution of the primal problem from the optimal solution of the dual problem. Clearly describe the result presented in the course that you use to do so.

Problem 5

Consider the following linear programming problem:

$$
\begin{aligned}
\max\ z = 2x_1 + x_2 + 3x_3 & \\
x_1 + x_2 + 3x_3 &\leq 6 \\
6x_1 + 3x_2 + 6x_3 &\leq 15 \\
x_1, x_2, x_3 &\geq 0
\end{aligned}
$$

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a) Express the problem in standard form.

b) Starting from the initial basic feasible solution where all the slack variables are in the basis, perform <u>one iteration</u> of the Simplex algorithm to bring the tableau in <u>canonical form associated to the vertex</u> $\tilde{\mathbf{x}} = (0, 5, 0, 1, 0)$.

Indicate the resulting tableau and the objective function value of the associated basic feasible solution. Also report the initial tableau and the main steps.

c) Is the basic feasible solution $\tilde{\mathbf{x}}$ optimal? Motivate your answer. If it is not optimal, indicate which variable must enter the basis and which variable must exit the basis.

d) Write the dual problem.

e) What can we say concerning the primal problem by just considering the dual problem? Motivate your answer.

Problem 6

Consider the following integer linear programming (ILP) problem:

$$\begin{array}{rll} \min & x_1 - 3x_2 & \\ s.t. & -2x_1 + 5x_2 & \leq 10 \\ & 6x_1 + 5x_2 & \leq 30 \\ & x_1, x_2 & \geq 0 \ \text{integer.} \end{array}$$

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a) Describe how the linear programming relaxation is used in the Branch-and-Bound method for solving ILP problems.

b) Find the optimal solution of the above ILP problem with the Branch-and-Bound method by solving graphically the linear programming relaxation of the subproblems. When it applies <u>consider first</u> the <u>subproblem with the best bound</u>.

Report all steps, namely the branching tree, the graphical solutions of all subproblems with the corresponding solutions and bounds.

c) Indicate the optimal solution found and its objective function value.

Problem 7

Consider the Integer Linear Programming (ILP) problem:

$$\begin{aligned} \max\ z = 3x_A + x_B & \\ x_A &\leq 3 \\ x_B &\leq 2 \\ 2x_A + 2x_B &\leq 7 \\ x_A, x_B &\geq 0 \text{ integer} \end{aligned}$$

with the following optimal tableau of the linear relaxation:

		x_A	x_B	s_1	s_2	s_3
	-19/2	0	0	- 2	0	-1/2
x_A	3	1	0	1	0	0
x_B	1/2	0	1	-1	0	1/2
s_2	3/2	0	0	1	1	-1/2

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a) Briefly describe the idea of the cutting plane method with Gomory fractional cutting planes for ILP.

b) Determine all the possible Gomory cutting planes for the optimal solution of the linear relaxation. What do you observe?

c) Express the Gomory cutting planes in terms of the decision variables x_A and x_B and draw them on the graphical representation of the feasible region of the ILP and the constraints.

d) To find the optimal integer solution, do we need further iterations of the cutting plane method?

Problem 8

A distribution company has to supply a single type of good to n clients. Let d_j denote the demand of client j, where $j \in J = \{1, \ldots, n\}$. The goods must be stocked in warehouses before being delivered to the clients. The warehouses can be located in m candidate sites. Let f_i denote the cost for opening a warehouse at the candidate site $i \in I$, with $i \in I = \{1, \ldots, m\}$, and b_i denote its maximum capacity. Let c_{ij} be the unit transportation cost from warehouse i to client j, and q_{ij} be the maximum quantity to be transported from warehouse i to client j. According to local regulations, it is forbidden to open warehouses that are too close to each other. Let l_{ih} be the distance between warehouse $i \in I$ and warehouse $h \in I$ and L_{min} be the minimum distance allowed between two distinct warehouses. The goal is to decide where to locate the warehouses and how to satisfy the client demands so as to minimize the total costs, i.e., the total opening costs plus the total transportation costs.

Using the variables

x_{ij}: amount of good to be transported from warehouse $i \in I$ to client $j \in J$

y_i: 1 if warehouse $i \in I$ is active, and 0 otherwise,

the problem can be formulated as follows:

$$
\begin{aligned}
\min \; & \sum_{i \in I, j \in J} c_{ij} x_{ij} + \sum_{i \in I} f_i y_i \\
\text{s.t.} \; & \sum_{j \in J} x_{ij} \leq b_i y_i && \forall i \in I \\
& \sum_{i \in I} x_{ij} \geq d_j && \forall j \in J \\
& y_i + y_h \leq 1 && \forall i \in I, \quad \forall h \in I, \quad l_{ih} < L_{min} \\
& 0 \leq x_{ij} \leq q_{ij} && \forall i \in I, j \in J \\
& y_i \in \{0, 1\} && \forall i \in I
\end{aligned}
$$

Translate the above model into the AMPL language: provide *on the back of the previous page* the complete .mod with all the declarations of sets, parameters, variables, constraints, and objective function.

Assume that the .dat file contains the values of the following parameters: n, m, L_{min}, d_j, f_i, b_i, c_{ij} and l_{ih}. Note that the .dat file does not specify the members of the sets J and I, but only the cardinality n of the set J and the cardinality m of the set I.

Problem 9

Suppose that three products with given demands d_i, with $i = 1, \ldots, 3$, have to be loaded onto a tanker. The tanker has five cargo holes and products cannot be blended into a cargo hole. The following table indicates the capacities k_j, with $j = 1, \ldots, 5$, of the five cargo holes.

Cargo hole	1	2	3	4	5
Capacity k_j (ton)	5400	5600	2200	1800	3400

All demands need not be delivered. But in case a demand is not delivered a penalty cost p_i, with $i = 1, \ldots, 3$, per ton must be paid. The following table indicates for each product the demands d_i, the maximum amount not delivered m_i and the penalty cost p_i per ton.

Product	Demand d_i (ton)	Max non-deliveries m_i (ton)	Penalty cost p_i (per ton)
91-octane	4500	800	0.5
95-octane	6000	1800	0.4
96-octane	6800	1000	0.3

The problem is to determine how to allocate the products to the cargo holes so as to minimize the penalty costs while respecting the loading constraints.

Give an integer linear programming formulation for the problem.

Decision variables:

Objective function:

Constraints:

Problem 10

A web service provider, which owns a set of m servers, receives a set of n requests. Each request $j \in \{1, \ldots, n\}$ consists in a software application to be run on a server. For each server i, with $i \in \{1, \ldots, m\}$, let M_i denote its memory capacity and C_i its computing capacity. For each request j the maximum memory consumption m_j and the maximum computing power consumption c_j are known. To guarantee a suitable level of service, the service provider must guarantee that, for each server, at most 90% of its memory capacity and computing capacity are used. Due to security reasons (in case of attacks or viruses), some pairs of applications cannot be run on the same server. For each pair of requests j and l, with $j \neq l$, a parameter a_{jl} indicates whether they are compatible: $a_{jl} = 1$ if requests j and l can be run on the same server, and $a_{jl} = 0$ otherwise.

The service provider has to decide how to allocate all the requests to the servers so as to minimize the number of servers used, while satisfying the suitable level of service and the compatibility constraints.

Give an integer linear programming formulation for the problem.

Decision variables:

Objective function:

Constraints:

Problem 11

A company has to supply a new type of product to n stores across a given area, and it may use m warehouses to do so.
Let

- p_i denote the maximum capacity of the i-th warehouse, for $1 \leq i \leq m$,
- r_j denote the demand of the j-th store, for $1 \leq j \leq n$,
- f_i denote the fixed cost for using warehouse i, for $1 \leq i \leq m$,
- c_{ij} denote the unit transportation cost from warehouse i to store j, for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Due to truck capacity at most k_{ij} units of product can be transported from warehouse i to store j, for $1 \leq i \leq m$ and $1 \leq j \leq n$.

To avoid excessive fragmentation of the delivery, each store must receive the requested product units from at most 3 warehouses. Moreover, at most 5 warehouses can be used.

The problem is to determine a supply plan which minimizes the total (fixed and transportation) costs, while satisfying all demands and other constraints.

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a) Give an integer linear programming formulation for the problem.

Decision variables:

Objective function:

Constraints:

b) Translate the above model into the AMPL language: provide the complete .mod with all the declarations of sets, parameters, variables, constraints, and objective function. Assume that the .dat file contains the values of the following parameters: n, m, p_i, r_j, f_i, $c_{i,j}$ and k_{ij} for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Problem 12

A company, which produces one type of electronic device, has to plan the production for an n-period time horizon.
For each period $t \in \{1, \ldots, n\}$, let

- p_t denote the unit production cost in period t,
- s_t denote the unit storage cost in period t,
- d_t denote the demand in period t.

The production plant has a regular maximum capacity of C units per period which can be increased by up to C_{add} units per period at a fixed extra cost of f_t and with a unit production cost of $1.15p_t$, for $t \in \{1, \ldots, n\}$. The electronic devices can be produced in advance and can be stored in a depot of capacity D to satisfy the demand of future periods.

For technical reasons, if the production is switched on in a given period it cannot be switched off the next period, in other words production has to occur for at least two consecutive periods. Moreover, the production cannot occur in more than k periods out of the n-period time horizon, with $2 \leq k \leq n$.

Finally the stock, which is equal to U units at the beginning of the planning horizon, must also be equal U units at the end of the n-th period.

The problem is to determine a production plan for the next n periods that minimizes the total costs, while satisfying the demand in each period and all other constraints.

Give an integer linear programming formulation for the problem, assuming that there exits a feasible solution for the given values of the parameters.

Decision variables:

Objective function:

Constraints: